

PHARM: MANAGEMENT OF TYPE 2 DIABETES MELLITUS

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Disclosures

I have no conflicts of interest to disclose

Learning Objectives

1. Explain the mechanism of action of the non-insulin therapies for diabetes
2. Describe the clinical use and efficacy of each class of drugs
3. Identify the most common side effects of each drug class and use adverse-effect profiles to guide drug selection in clinical scenarios.
4. Identify the contraindications for each drug class
5. Formulate a therapeutic management plan to treat type 2 diabetes

Why know Diabetes?

38.4 million people have diabetes (11.6% of US population) (2021)

8.7 million adults (3.4% of the population) who met lab criteria for diabetes were not aware of or did not report having diabetes (undiagnosed).

8th leading cause of death in 2021

Heart disease and stroke are 2x more likely in diabetics

Leading cause of **ESRD (renal failure)**

Leading cause of new cases of **blindness** in adults

Leading cause of non-traumatic lower limb **amputations**

\$413 Billion = total direct and indirect estimated costs of diagnosed diabetes
(per the CDC, in 2024)

If Blood Sugars are Controlled, this Dramatically reduces the Risk for Microvascular Complications



DCCT- Type 1 Diabetics

Compared patients who intensively managed their glucose levels (A1c of 7.2) to those who did usual management (A1c of 9.2) :

Lower incidence of retinopathy, microalbuminuria, decreased progression of existing retinopathy and nephropathy, and a reduced incidence of neuropathy

UKPDS- Type 2 Diabetics

Compared A1C 7.0 % (intensive group) vs 7.9% (conventional group):

25 % risk reduction in microvascular disease in intensive group. No evidence of a threshold effect for A1C;

1% fall in A1C associated with 35% reduction in microvascular endpoints

Prevention of Type 2 Diabetes

Great evidence for screening! Effective preventive strategies: Weight loss, lifestyle changes

- Screen for prediabetes and/or type 2 DM in asymptomatic people: considered in adults of any age with overweight or obesity (BMI ≥ 25 kg/m² or ≥ 23 kg/m² in Asian Americans) who have one or more risk factors
- Adults: screening at age 35 (ADA) or age 40 (USPSTF)

The Diabetes Prevention Program

- Refer patients with prediabetes to intensive behavioral lifestyle intervention program, achieve and maintain 7% loss of initial body weight, increase moderate-intensity physical activity 150 min/ week (such as brisk walking)
- Can reduce the risk by 58%!

Type 2 Diabetes

Insulin resistance in the muscle and liver and **β-cell failure** represent the core pathophysiologic defects in type 2 diabetes

Other defects that also contribute to hyperglycemia:

- the fat cell (accelerated lipolysis)
- gastrointestinal tract (incretin deficiency/resistance)
- α-cell (hyperglucagonemia),
- kidney (increased glucose reabsorption)
- brain (insulin resistance)

....all play important roles in the development of glucose intolerance in type 2 diabetic individuals, and all important to understand the mechanisms of action and selection of the pharmacologic agents we use for Type 2 diabetes

Management Strategies

Today we are focusing on NON- Insulin agents, but we also need to pay attention to co-morbidities

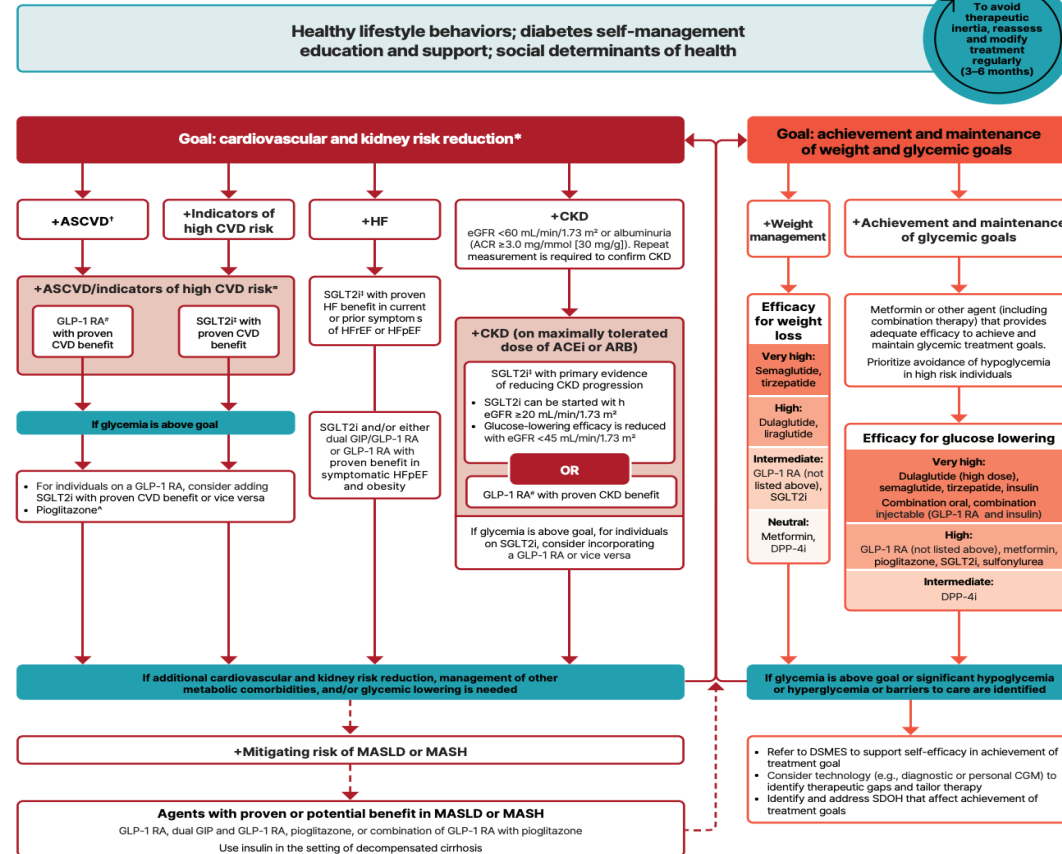
Patients with T2DM are at least 2 times more likely to develop cardiovascular (CV) diseases due to increased risk factors, such as hypertension, obesity, and hyperlipidemia

Don't forget that these other risk factors should also be addressed (HTN control, statin use, weight loss, smoking cessation)

Diabetes Care 2026; 49 (Suppl 1, Sect 9)

Use of glucose-lowering medications in the management of type 2 diabetes

(For recommendations for specific conditions, including non-glucose-lowering medications, refer to pertinent sections)



* In people with HF, CKD, established CVD, or multiple risk factors for CVD, the decision to use a GLP-1 RA or SGLT2i with proven benefit should be made irrespective of attainment of glycemic goal.

† ASCVD: Defined differently across CVOTs but all included individuals with established CVD (e.g., MI, stroke, and arterial revascularization procedure) and variably included conditions such as transient ischemic attack, unstable angina, amputation, and symptomatic or asymptomatic coronary artery disease. Indicators of high risk: While definitions vary, most comprise ≥55 years of age with two or more additional risk factors (including obesity, hypertension, smoking, dyslipidemia, or albuminuria).

‡ A strong recommendation is warranted for people with CVD and a weaker recommendation for those with indicators of high risk CVD. Moreover, a higher absolute risk reduction and thus lower numbers needed to treat are seen at higher levels of baseline risk and should be factored into the shared decision-making process. See text for details.

For GLP-1 RAs, CVOTs demonstrate their efficacy in reducing composite MACE, CV death, all-cause mortality, MI, stroke, and kidney end points in individuals with T2D with established or high risk of CVD. One kidney outcome trial demonstrated benefit in reducing persistent eGFR reduction and CV death for a GLP-1 RA in individuals with CKD and T2D.

‡ For SGLT2is, CV and kidney outcomes trials demonstrate their efficacy in reducing the risks of composite MACE, CV death, all-cause mortality, MI, HF, and kidney outcomes in individuals with T2D and established or high risk of CVD.

^ Low-dose pioglitazone may be better tolerated and similarly effective as higher doses.

Figure 9.4—Use of glucose-lowering medications in the management of type 2 diabetes. The left side of the algorithm prioritizes mitigation of diabetes-related complications and end-organ effects, while the right side addresses weight and glucose management goals. ACEi, angiotensin-converting enzyme inhibitor; ACR, albumin-to-creatinine ratio; ARB, angiotensin receptor blocker; ASCVD, atherosclerotic cardiovascular disease; CGM, continuous glucose monitoring; CKD, chronic kidney disease; CV, cardiovascular; CVD, cardiovascular disease; CVOT, cardiovascular outcomes trial; DPP-4i, dipeptidyl peptidase 4 inhibitor; DSMES, diabetes self-management education and support; eGFR, estimated glomerular filtration rate; GLP-1 RA, glucagon-like peptide 1 receptor agonist; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; HFrEF, hospitalization for heart failure; MACE, major adverse cardiovascular events; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease; MI, myocardial infarction; SDOH, social determinants of health; SGLT2i, sodium–glucose cotransporter 2 inhibitor; T2D, type 2 diabetes. Adapted from Davies et al. (90).

Healthy lifestyle behaviors; diabetes self-management education and support; social determinants of health

To avoid therapeutic inertia, reassess and modify treatment regularly (3–6 months)

Goal: cardiovascular and kidney risk reduction*

Goal: achievement and maintenance of weight and glycemic goals

1. Lifestyle changes, Diabetes Education and Support, SDOH
2. Check comorbidities or Risk level for Cardio-renal disease
3. Glycemic and weight management goals

Lifestyle Modification & Weight Loss

MNT (Medical Nutrition Therapy) delivered by a registered dietitian is associated with A1C decreases of: 0.3–2%

- A variety of eating patterns are acceptable (Mediterranean, Dietary Approaches to Stop Hypertension (DASH) , plant-based diets, portion control, reducing high glycemic index foods)

Physical activity -150 min or more of moderate-to-vigorous intensity aerobic activity per week (most adults)

- Shorter durations (75min/ week) of vigorous-intensity/ interval training may be sufficient for younger and more physically fit individuals
- Prolonged sitting should be interrupted every 30 min

Selecting Therapies

Goal: cardiovascular and kidney risk reduction*

+ASCVD[†]

**+Indicators of
high CVD risk**

+HF

+CKD

eGFR <60 mL/min/1.73 m² or albuminuria
(ACR ≥3.0 mg/mmol [30 mg/g]). Repeat
measurement is required to confirm CKD

Cardiorenal risk reduction in high- risk patients

CAD/ ASCVD, CHF, CKD?

As we go through pharmaceutical classes-
be mindful of EXTRAGLYCEMIC benefits

Glycemic Management

Minimize Weight Gain/ Promote Weight loss

Consider Cost and Access

Minimize Hypoglycemia

**Goal: achievement and maintenance
of weight and glycemic goals**

**+Weight
management**

**+Achievement and maintenance
of glycemic goals**

**Efficacy
for weight
loss**

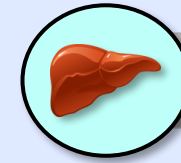
Very high:
Semaglutide,
tirzepatide

High:

Metformin or other agent (including
combination therapy) that provides
adequate efficacy to achieve and
maintain glycemic treatment goals.

Prioritize avoidance of hypoglycemia
in high risk individuals

Biguanide: Metformin



Glucose
production

TRADITIONALLY, FIRST LINE AGENT –we usually start metformin **at diagnosis** of Type 2 diabetes

- Weight neutral or modest weight loss
- Does not cause hypoglycemia
- Does not have adverse cardiovascular effects, and it appears to decrease cardiovascular events in certain populations
- Inexpensive!

Mechanism of Action: decrease hepatic glucose output/ inhibits gluconeogenesis. It may increase insulin-mediated glucose utilization in peripheral tissues

A1c lowering (up to /approx. 1.5%)

Metformin

Metabolism

Not metabolized and is **excreted by the kidneys as the active compound**

Safe to use with estimated glomerular filtration rates (eGFR) between 45-60 mL/min per 1.73 m²

eGFR 30 -44 mL/min per 1.73 m²: cautious use (max dose 1000 mg/d)

GFR < 30 mL/min per 1.73 m²: contraindicated

Should temporarily be held on day of radiocontrast administration & restarted 1-days later after confirmation that renal function has not deteriorated

Renal function should be checked at least annually in patients on metformin.
Use lower doses in the elderly who may have limited renal reserve

Metformin

Side Effects

Gastrointestinal: Most common (up to 20%) diarrhea, abdominal discomfort, nausea (taking with food and slowly up titration of the dose can help alleviate)

B12 deficiency (after many years of use) – it may interfere with absorption of vitamin B₁₂–intrinsic factor complex in the terminal ileum (periodic screening for B₁₂ deficiency should be considered, especially in patients with peripheral neuropathy)

Lactic Acidosis Rare but dangerous side effect. As a consequence of Metformin blocking gluconeogenesis, it may impair the hepatic metabolism of lactic acid

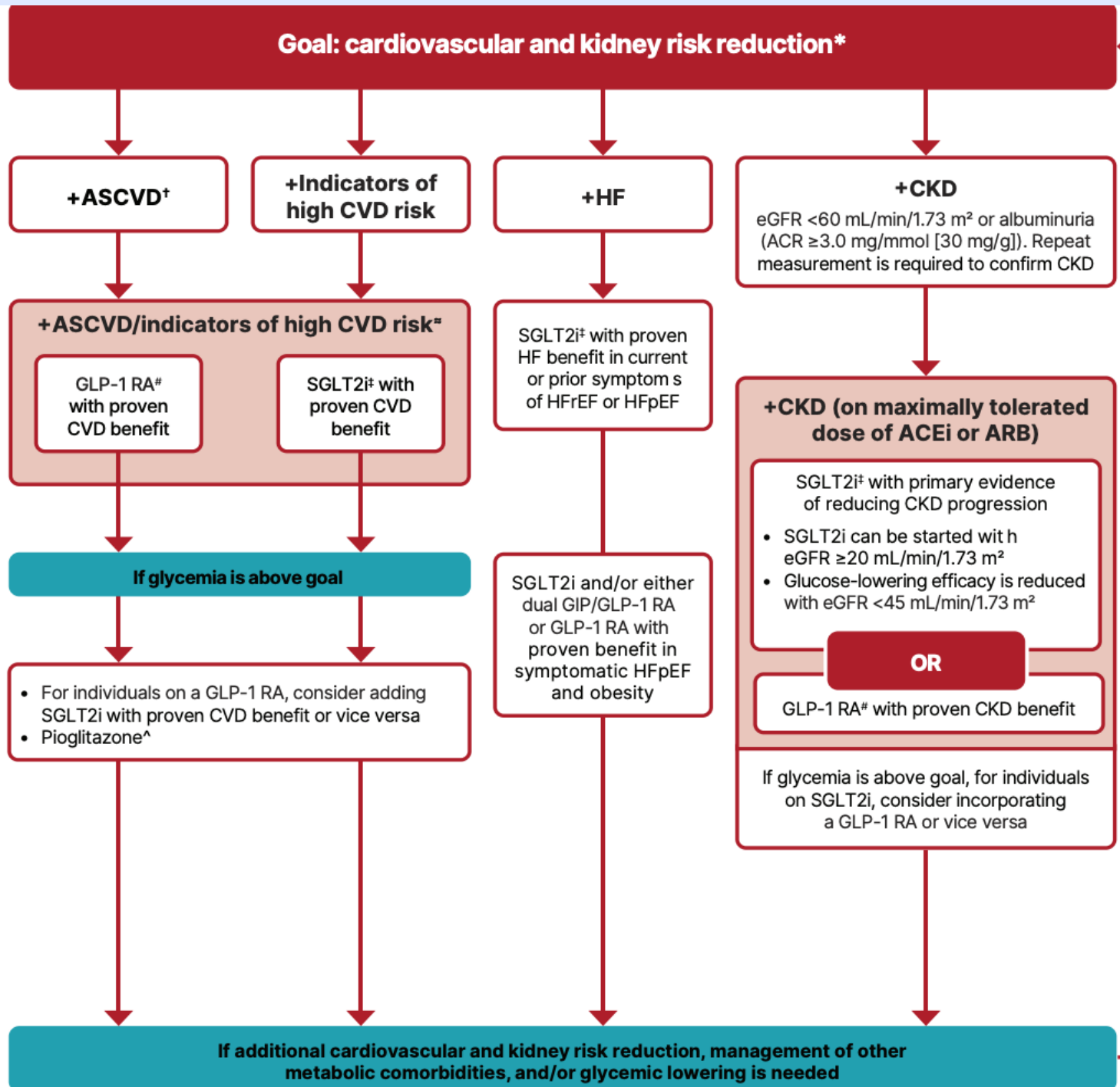
Almost all reported cases have involved patients with associated risk factors that should have contraindicated its use; conditions of tissue hypoxia when there is increased production of lactic acid (ex. heart failure, shock , sepsis, active liver disease, active alcohol abuse) and in renal failure when there is decreased clearance of metformin

Selecting Therapies

Prior guidelines (2022) advised metformin at diagnosis, and most often you will see that continuing

However, guidelines reinforce that you also need to evaluate **Comorbidities** and this can guide you to specific medications

- Does your patient have ASCVD, CHF, or CKD?
- Does your patient have high risk for these
 - ≥ 55 with 2 or more RF (obesity HTN, smoking, dyslipidemia, or albuminuria?)



Incretins

Gut Hormones which normally help with our glucose control

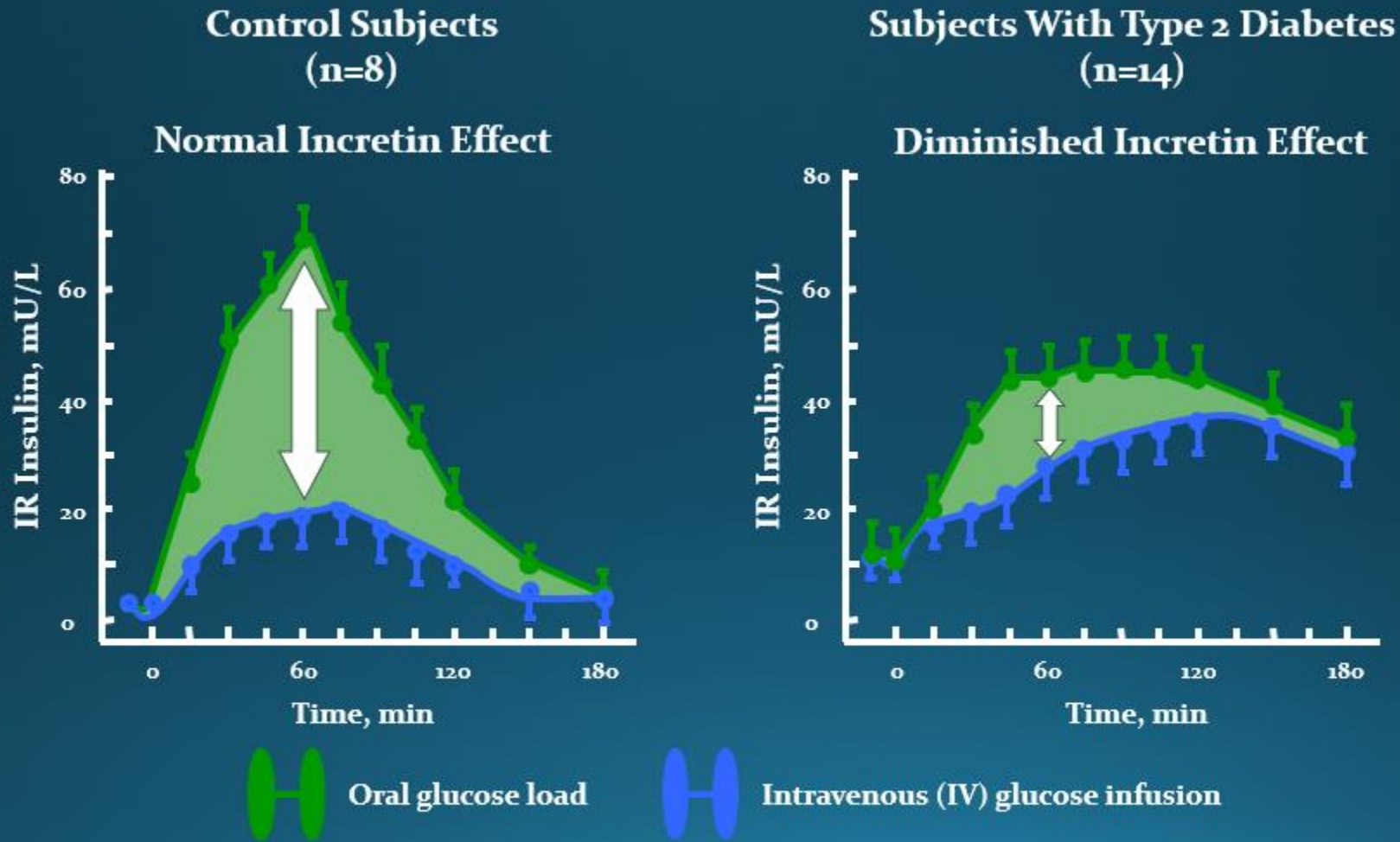
- Stimulation of insulin secretion only when blood sugar is high / fed state
- Lowers both post meal and fasting glucose
- Suppresses inappropriately elevated glucagon post-meal
- Reductions in weight

Diabetics -demonstrated to exhibit almost total loss of incretin effect

GLP1: Glucagon Like Peptide 1

GIP: glucose-dependent insulinotropic peptide

What is the Incretin Effect?



Adapted with permission from Nauck M et al. *Diabetologia* 1986;29:46-52. Copyright © 1986 Springer-Verlag.

Goal: cardiovascular and kidney risk reduction*

+ASCVD[†]

+Indicators of high CVD risk

+HF

+CKD

eGFR <60 mL/min/1.73 m² or albuminuria (ACR ≥3.0 mg/mmol [30 mg/g]). Repeat measurement is required to confirm CKD

+ASCVD/indicators of high CVD risk*

GLP-1 RA[#] with proven CVD benefit

SGLT2i[‡] with proven CVD benefit

SGLT2i[‡] with proven HF benefit in current or prior symptoms of HFrEF or HFpEF

+CKD (on maximally tolerated dose of ACEi or ARB)

SGLT2i[‡] with primary evidence of reducing CKD progression

- SGLT2i can be started with eGFR ≥20 mL/min/1.73 m²
- Glucose-lowering efficacy is reduced with eGFR <45 mL/min/1.73 m²

OR

GLP-1 RA[#] with proven CKD benefit

If glycemia is above goal, for individuals on SGLT2i, consider incorporating a GLP-1 RA or vice versa

If glycemia is above goal

- For individuals on a GLP-1 RA, consider adding SGLT2i with proven CVD benefit or vice versa
- Pioglitazone[^]

SGLT2i and/or either dual GIP/GLP-1 RA or GLP-1 RA with proven benefit in symptomatic HFpEF and obesity

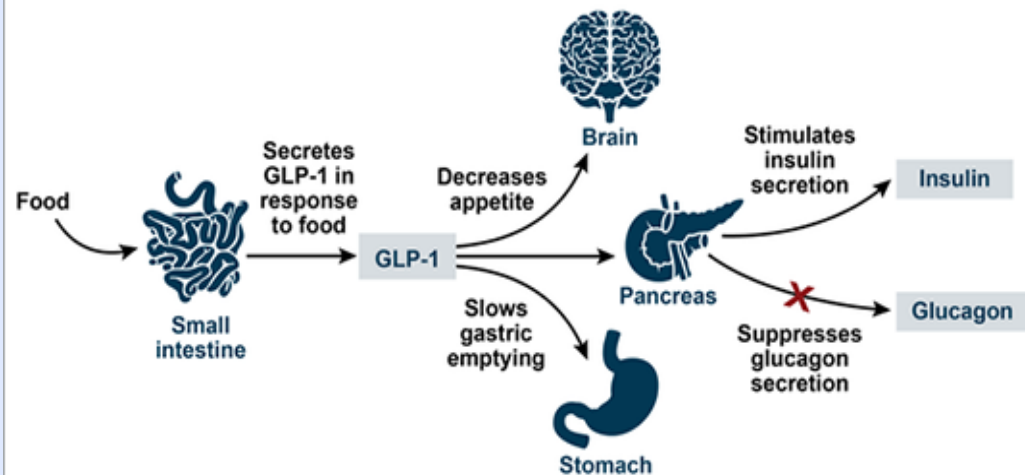
If additional cardiovascular and kidney risk reduction, management of other metabolic comorbidities, and/or glycemic lowering is needed

GLP1- Glucagon Like Peptide 1

GLP-1

- Is released from L cells in ileum and colon^{1,2}
- Stimulates insulin response from beta cells in a glucose-dependent manner¹
- Inhibits glucagon secretion from alpha cells in a glucose-dependent manner¹
- Inhibits gastric emptying^{1,2}
- Reduces food intake and body weight²

GLP-1 RAs *Mechanism of Action*



Meier JJ. *Nat Rev Endocrinol.* 2012;8:728-742.

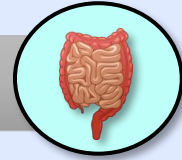
GLP-1=glucagon-like peptide-1; GIP=glucose-dependent insulintropic peptide.

1. Meier JJ et al. *Best Pract Res Clin Endocrinol Metab.* 2004;18:587-606.

2. Drucker DJ. *Diabetes Care.* 2003;26:2929-2940

GLP-1 Receptor Agonists (GLP1RA)

Incretin
effect



- May be most known for weight loss effects!
- Do not cause hypoglycemia
- Next to insulin, the strongest for A1c lowering
- + Cardiovascular benefit in RCT's
- Can be expensive

Mechanism of Action: Synthetic glucagon-like peptide-1 (GLP-1) receptor agonists bind to the GLP-1 receptor and exhibit the effects of GLP1. They are resistant to degradation by the enzyme dipeptidyl peptidase-4 (DPP-4) and therefore have a longer half-life, facilitating clinical use

GLP₁RA's ("tides")

All are injectable except one....

Daily Use

Exenatide

Liraglutide*

Semaglutide Oral*

Weekly Use

Exenatide extended
release

Dulaglutide*

Semaglutide* ^

*Proven to reduce MACE (Major Adverse Cardiac Events):
Death from cardiovascular cause, nonfatal MI, nonfatal stroke

^Proven to reduce new or worsening nephropathy
Decline in GFR, progression of CKD

GLP-1RA (“tides”)

Clinical effects

- All injectable except for oral semaglutide
- **Glycemic efficacy** - In comparison with placebo, all GLP-1 receptor agonists reduced glycated hemoglobin (A1C) by approximately 1 percentage point or more
- **Weight loss** – Patients lose at least 2-3 kg in most trials, some trials greater weight loss benefit

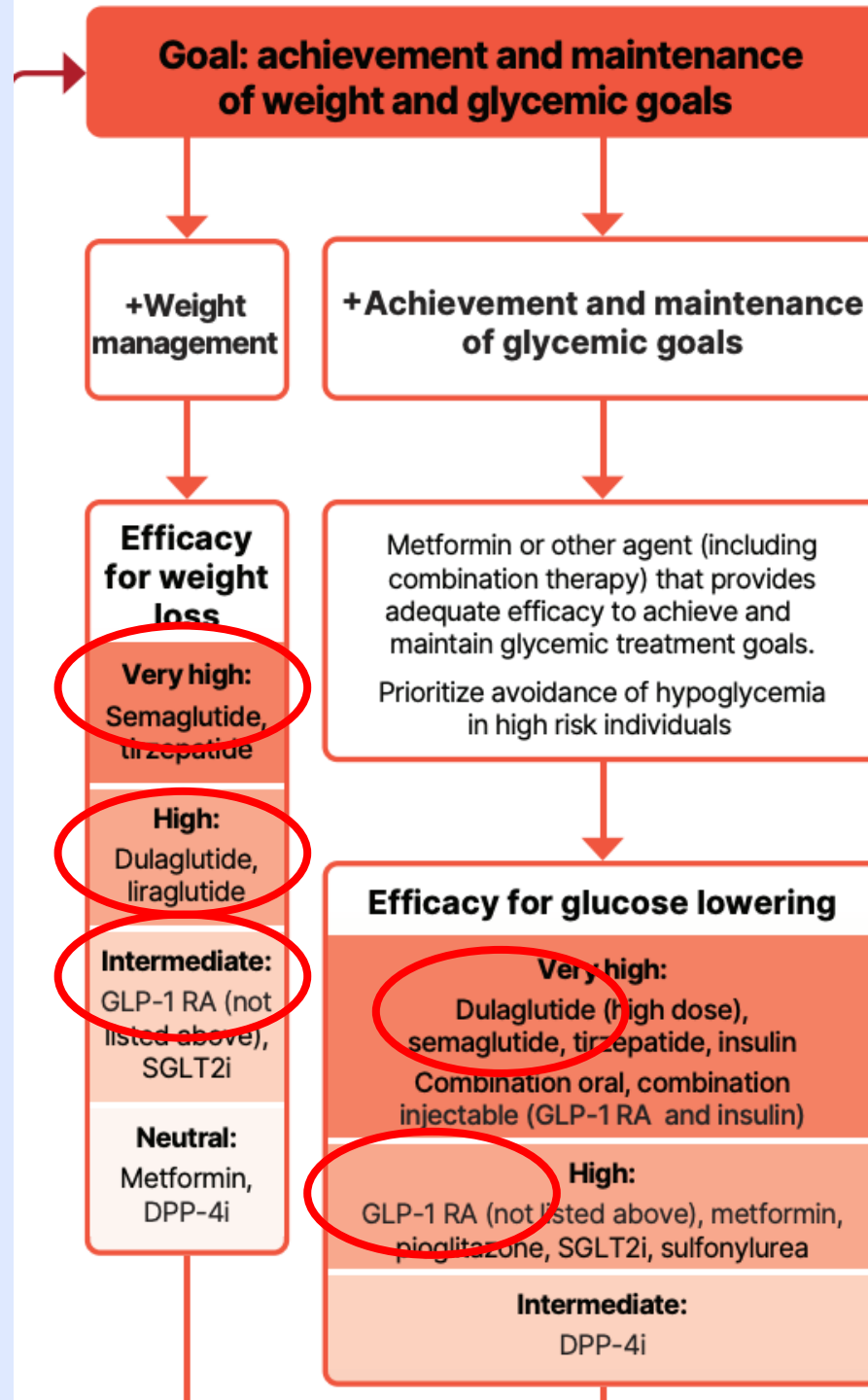
GLP-1 agonists (“tides”)

Adverse effects

Gastrointestinal - nausea, abdominal fullness, vomiting, and diarrhea in 10-50% patients (remember they slow gastric emptying).

Pancreas — Acute pancreatitis has been reported *in association* with GLP-1 agonist treatment. There is a warning against GLP-1 receptor agonist use in patients with a history of pancreatitis.

GLP-1 agonists are contraindicated in patients with a personal or family history of **medullary thyroid cancer** or multiple endocrine neoplasia 2A or 2B. In rodent studies hyperplasia of thyroid C cells was noted, and therefore a potential for the risk of medullary thyroid cancer.



Selecting Therapies

So we started with lifestyle, metformin....

If ASCVD, need agent with high efficacy for glucose lowering, or high efficacy for weight management goals

Next step may be a GLP1RA

(cost & patient preferences have to be considered)

If ASCVD, CHF or CKD, need to minimize hypoglycemia

Next step might have been the **SGLT2-inhibitor class** (SGLT2i)...

Goal: cardiovascular and kidney risk reduction*

+ASCVD[†]

+Indicators of high CVD risk

+HF

+CKD

eGFR <60 mL/min/1.73 m² or albuminuria (ACR ≥3.0 mg/mmol [30 mg/g]). Repeat measurement is required to confirm CKD

+ASCVD/indicators of high CVD risk*

GLP-1 RA[#] with proven CVD benefit

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- For individuals on a GLP-1 RA, consider adding SGLT2i with proven CVD benefit or vice versa
- Pioglitazone[^]

SGLT2i and/or either dual GIP/GLP-1 RA or GLP-1 RA with proven benefit in symptomatic HFpEF and obesity

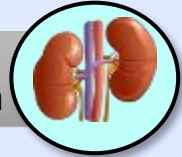
If glycemia is above goal, for individuals on SGLT2i, consider incorporating a GLP-1 RA or vice versa

If additional cardiovascular and kidney risk reduction, management of other metabolic comorbidities, and/or glycemic lowering is needed

Sodium-Glucose Transporter 2 Inhibitors

SGLT2i (“flozins”)

Glucose
Reabsorption



- 1886, German physician Joseph von Mering demonstrated that the ingestion of high doses of phlorizin (from the bark of apple trees) caused people to have glucosuria
- Modest weight loss benefit
- No risk of hypoglycemia
- +Cardiovascular and renal benefit in RCT's
- Can be expensive
- Once daily pills

Canagliflozin

Dapagliflozin

Empagliflozin

Ertugliflozin

Sodium-Glucose Transporter 2 Inhibitors SGLT2i

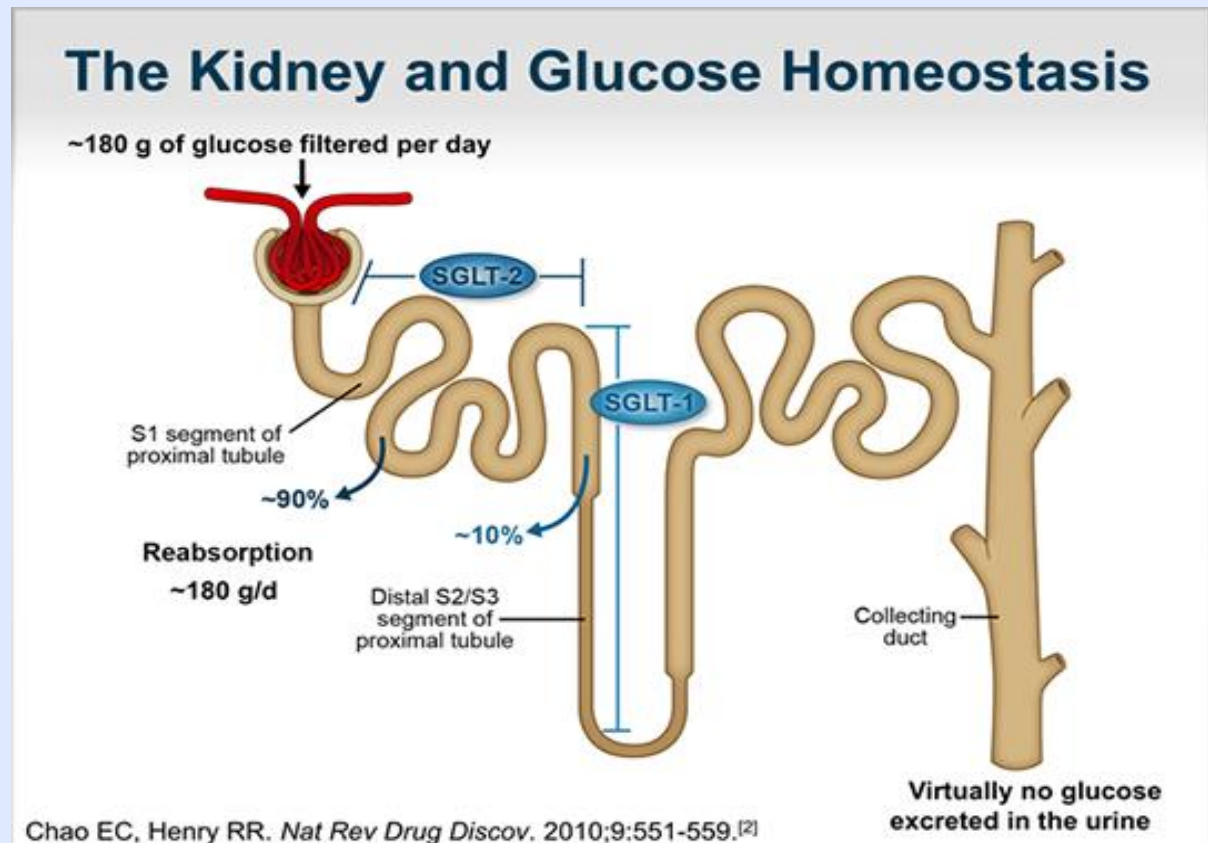
Mechanism of Action/ Metabolism

Inhibits sodium glucose transporter 2, in the proximal tubule

Inhibition reduces reabsorption of filtered glucose in tubules

Increases Urinary Excretion of Glucose

Metabolism: Liver, with renal/ feces elimination



SGLT2i

Clinical effects

- **Glycemic efficacy**— mean reductions in A1C compared with placebo 0.4 to 1.1% .
- **Weight loss**- SGLT2 inhibitors decrease weight (2 to 3 kg)
- **Cardiovascular effects** -associated with decreased cardiovascular events (decrease in MACE) and decreased risk for heart failure and heart failure hospitalization. They also lower blood pressure 3-5 mmHg.
- **Renal Effects**- These agents are associated with decrease progression of kidney disease.

SGLT2i

Side effects

- **Genitourinary tract** — There is an approximate 2-4x increased incidence of vulvovaginal candidiasis and urinary tract infections.
- **Necrotizing Fasciitis** of the perineum (fourniers gangrene)
- **Hypotension/ volume depletion** — SGLT2 inhibitors cause an osmotic diuresis and intravascular volume contraction.
- **Acute kidney injury** — There have been post marketing reports of acute kidney injury, some requiring hospitalization and dialysis.
- **Diabetic ketoacidosis** — "Euglycemic" (usually meaning plasma glucose <250 mg/dL) diabetic ketoacidosis has been reported in patients with type 2 diabetes taking SGLT2 inhibitors.

SGLT2i

Contraindications/ precautions

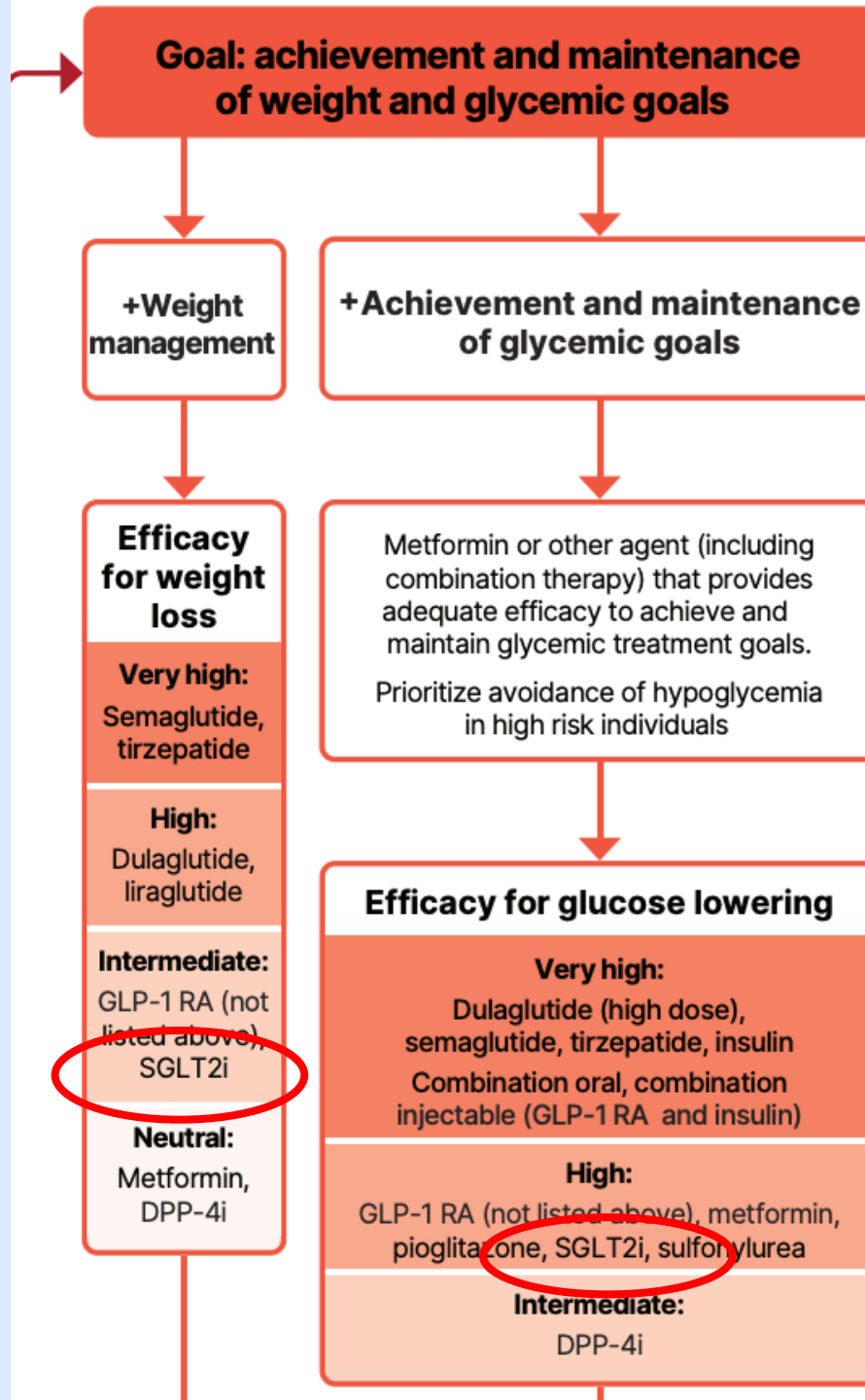
SGLT2 inhibitors should not be used in patients with:

- Type 1 diabetes
- Ketosis prone diabetes
- Patient on Dialysis

These agents should be held for surgery, NPO states and dehydration to reduce the risk of euglycemic DKA

These agents are also used for their renal benefit

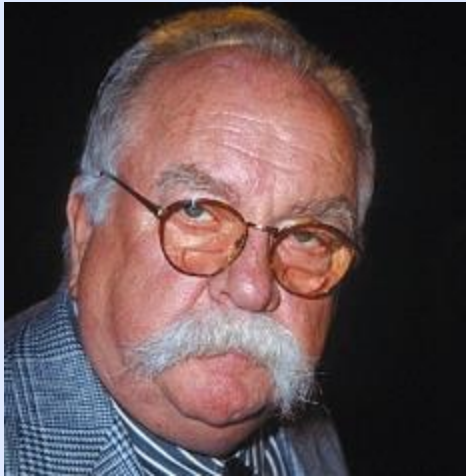
- They each have GFR recommendations (many are now ok to use down to an eGFR 30 mL/min/1.73 m)
- But each one has specific parameters (more data coming out- fast moving area of research)



Management Strategies

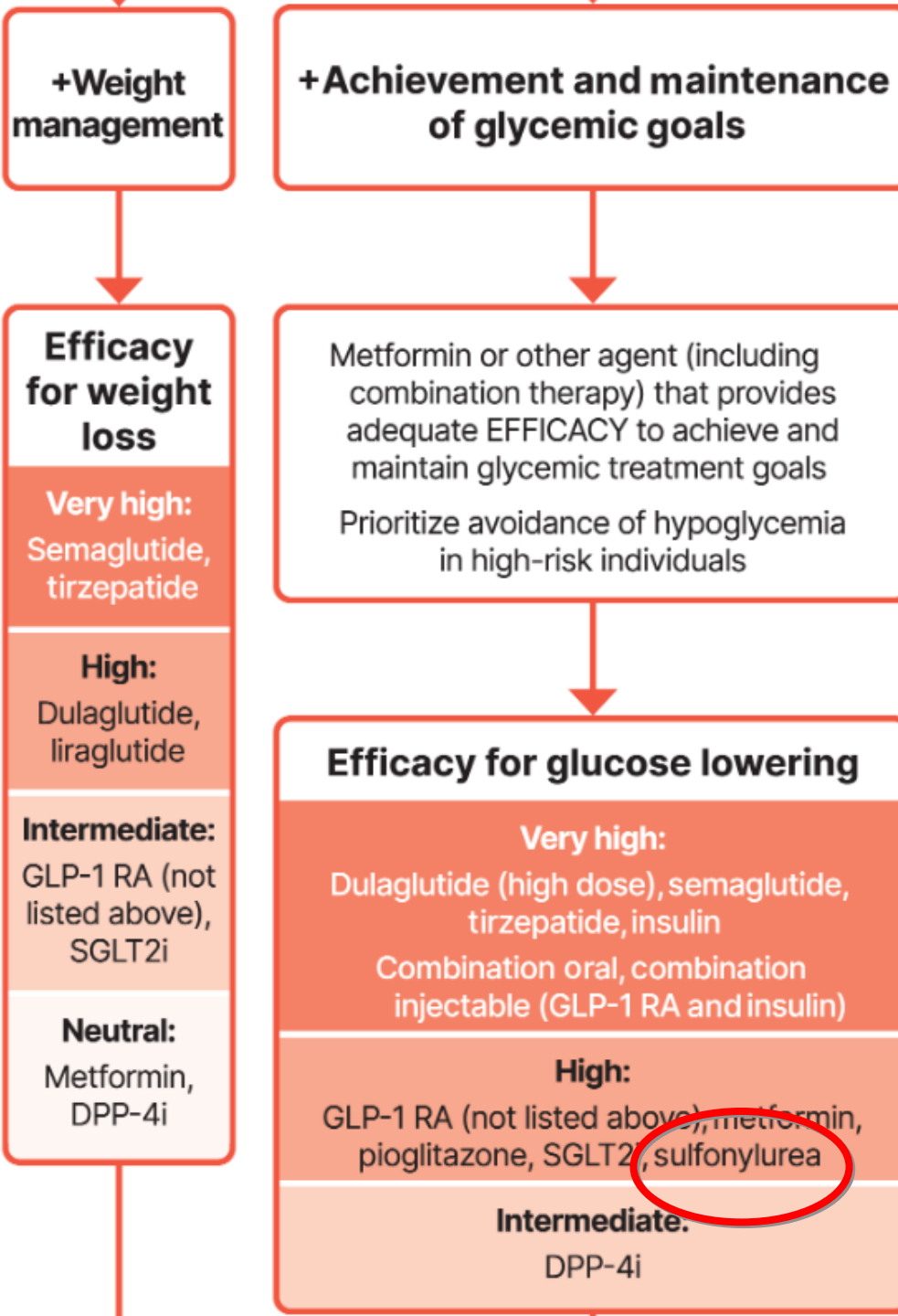
- Glucagon-like peptide-1 (GLP-1) receptor agonists and sodium-glucose co-transporter 2 (SGLT2) inhibitors can be initiated **independent of A1C and metformin therapy**, particularly for the individuals at high risk for heart failure, CKD (chronic kidney disease), and/or ASCVD (atherosclerotic cardiovascular disease)
- Both classes offer cardiovascular benefits and lower all-cause mortality while promoting glycemic control and weight loss

Famous Type 2 Diabetics



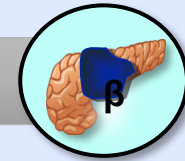
Selecting Therapies

- We did lifestyle changes, metformin and went over the glycemic and non-glycemic benefits of 2 classes, the GLP1RA and SGLT2i.
- Next, lets review agents that are LOW cost
- And then some less commonly used diabetic agents



Sulfonylureas

Insulin
secretion

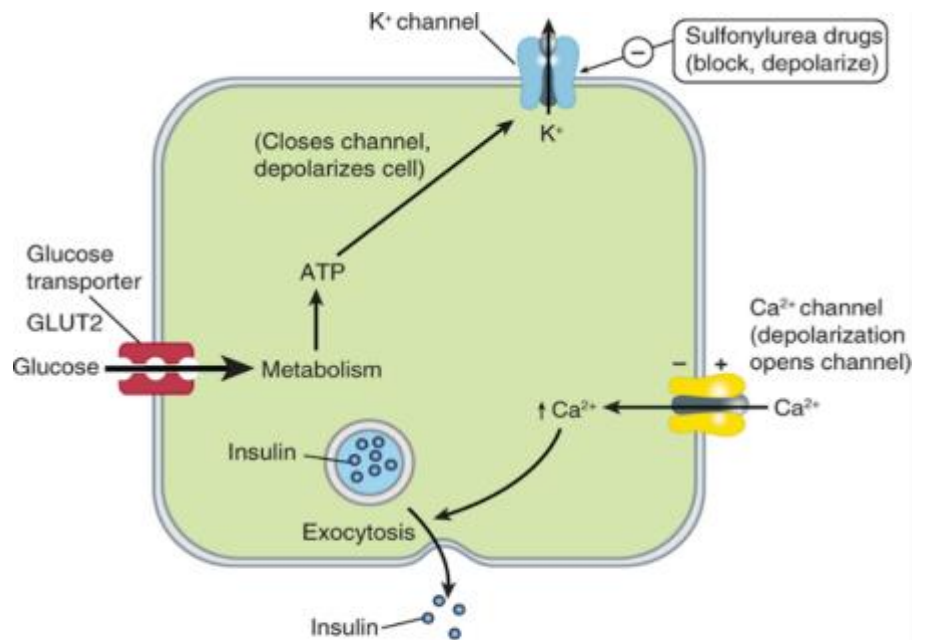


- Mechanism of Action: Enhances *Insulin secretion* by closing K-ATP channels on Beta cell membranes
- Known for their risk of **Hypoglycemia**
- Associated with **Weight gain**
- **Low cost**
- No known CVD benefits
- Usually lower A1c approx. 1-2%

Glyburide

Glipizide

Glimepiride



Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition
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Sulfonylureas

Metabolism/ Side effects/ Risks

Metabolized by the liver to weakly active or inactive metabolites, which are excreted by the kidney and partly excreted in the bile

- **Glyburide** is metabolized into products with lower hypoglycemic activity
- **Glipizide** and **glimepiride** have inactive metabolites

Side Effects

- **Hypoglycemia** is the most common. Conditions that increase the risk of hypoglycemia: renal or hepatic function, exercise, missed meals, ethanol ingestion
- **Weight gain**
- **Less common**: dizziness, headache, skin reactions (rash, pruritis)
- **Sulfonamide allergy**: chemical similarities between sulfonamides and sulfonylureas. Use of sulfonylureas is not specifically contraindicated in patients with sulfonamide allergy, however a risk of a cross-reaction exists. Do not use in patients in whom a previous reaction has been severe.



Hypoglycemia (in real life and the movies)



Symptoms

Sweats, shakes, heart palpitations, hunger, confusion

Long standing insulin treated diabetic with frequent hypoglycemia may lose warning symptoms

Severe hypoglycemia: seizure, death!

Treatment

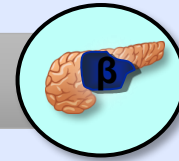
SIMPLE carbohydrate (JUICE, glucose tablets)

DO NOT eat fats/ proteins (**No candy bars, cookies, etc.**)

Glucagon injection

Meglitinides

Insulin
secretion



- Similar to SU: Mechanism of Action: regulate adenosine triphosphate (ATP)-sensitive potassium channels (K-ATP channels) in beta cells to increase insulin secretion.
 - Structurally different than sulfonylureas, exert effects via different pancreatic beta cell receptors
- Rapid onset and short duration of action. Administered with meals to reduce postprandial hyperglycemia
- Can cause Hypoglycemia
- More expensive than sulfonylureas, can be used if allergy to sulfonylurea medications
- No CVD benefit

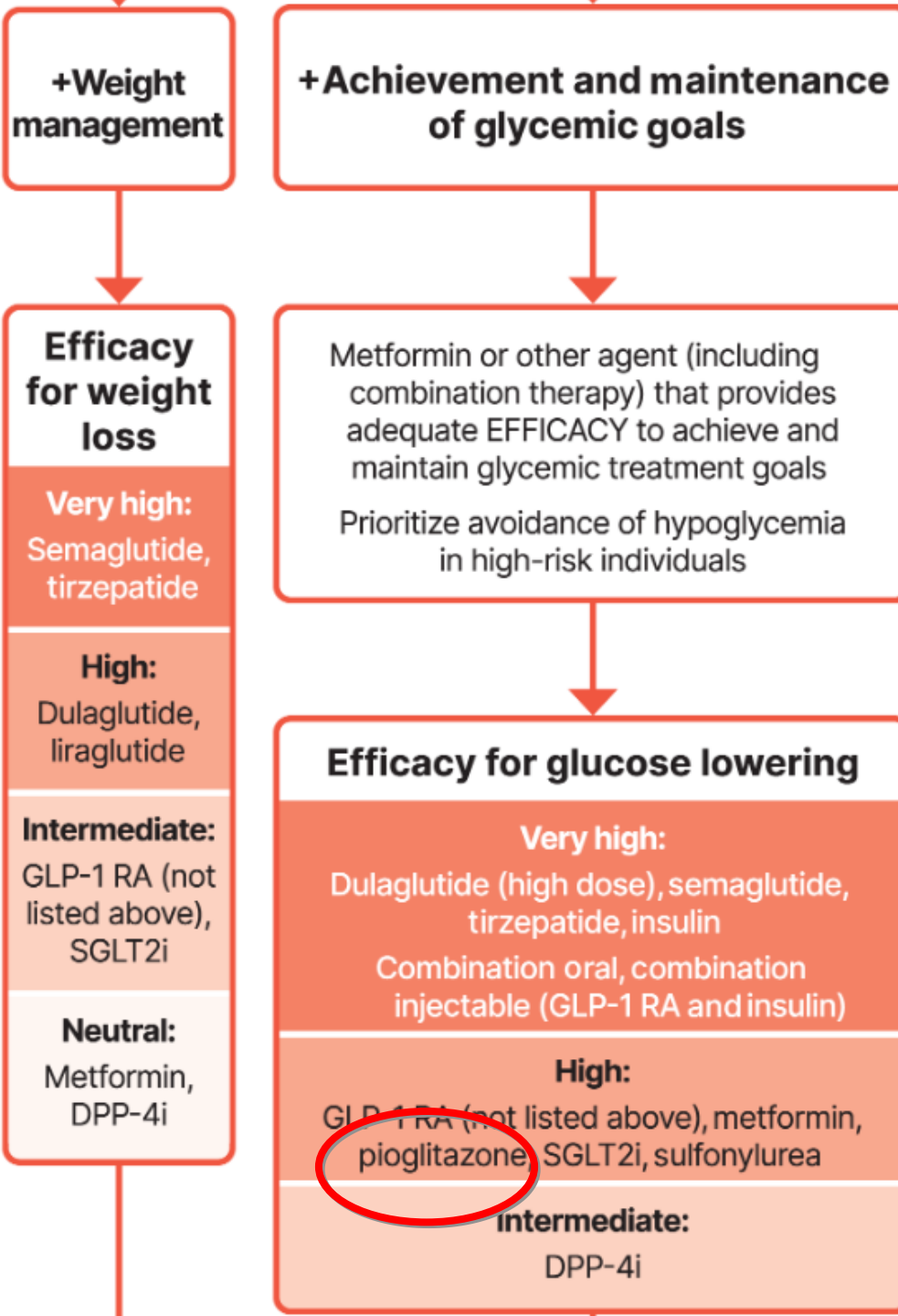
Repaglinide

Nateglinide

Meglitinides

Metabolism

- Repaglinide is principally metabolized by the liver, <10% renally excreted. Thus, it can be used more safely in patients with chronic kidney disease.
- Nateglinide is hepatically metabolized, with renal excretion of active metabolites. Use with caution in patients with chronic liver or renal disease
- These agents are not as commonly used



Thiazolidenediones (“TZD’s”)

Glucose
uptake



- Decrease insulin resistance, “insulin sensitizers”
- May be best known for side effects: **fluid retention**, possible precipitation or worsening of **congestive heart failure** and a meta-analysis showing possible increased **CVD**
- Associated with Weight Gain
- Inexpensive

Popular from late 1990’s- 2007, not commonly used now due to their side effect profile and since many other classes of medications became available

Pioglitazone

Rosiglitazone

Thiazolidinediones

Mechanism of Action

Insulin Sensitizers

- Ligands of **peroxisome proliferator-activated receptor gamma (PPAR- γ)** in **muscle, fat and liver**, part of the steroid and thyroid superfamily of nuclear receptors
- Modulates the expression of the genes involved in lipid and glucose metabolism, insulin signal transduction, and adipocyte and other tissue differentiation
- May increase expression of genes involved in promoting lipid storage and decrease expression of genes associated with inflammation, such as interleukin (IL)-6

Potential benefit- MASLD or MASH

TZD's

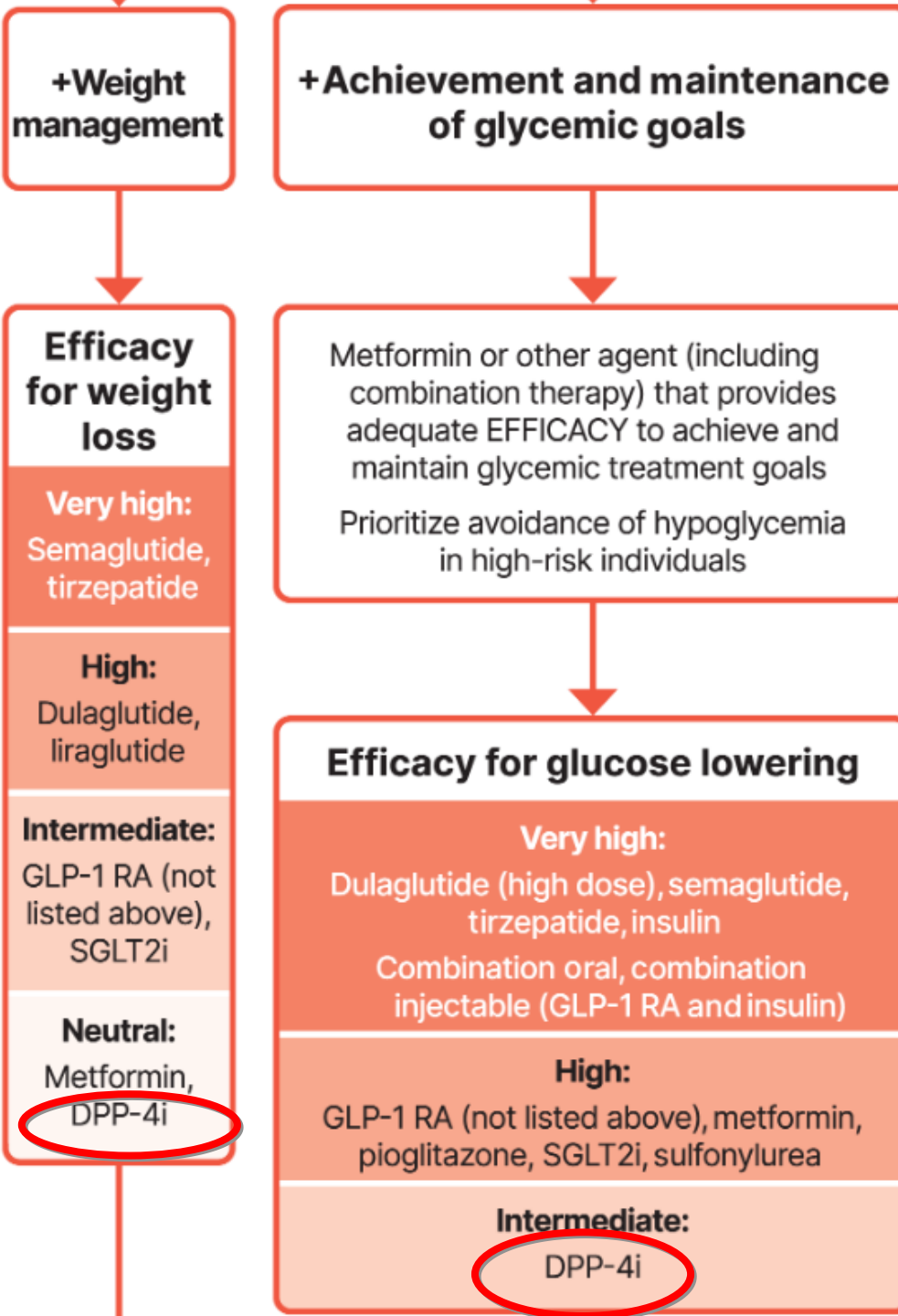
Cardiovascular effects/ metabolism

- **TZDs increase the risk of heart failure**
- The effect of TZDs on cardiovascular events and outcomes may depend upon the specificity of PPAR activation. Since pioglitazone effects both PPAR γ and PPAR α receptors, it has a more favorable effect on lipids.
- May have anti-inflammatory, antithrombotic and anti-atherogenic properties. They have been shown to improve a number of cardiovascular risk factors, and Pioglitazone has been associated with lower risk of myocardial infarction.
- **Metabolism** –metabolized by the liver. The Biologic half life is much longer than pharmacologic half – life (can take weeks to work and weeks for their effects to resolve).

TZD's

Other Side effects

- **Weight gain**
- **Skeletal fractures** — There is an increasing body of evidence suggesting that thiazolidinediones decrease bone density and increase fracture risk, particularly in women.
- **Bladder cancer** — concern that pioglitazone use increases the risk of bladder cancer.
- **Hepatotoxicity** - severe hepatic toxicity has occasionally been reported in patients receiving TZD's



DPP-4 Inhibitors (“gliptins”)

Incretin
effect



- Well tolerated pills, but weaker A1c lowering (A1c reductions (most 0.4-0.9%))
- Do not cause hypoglycemia
- Weight neutral
- No evidence of CVD benefit
- Can be expensive

Sitagliptin

Linagliptin

Saxagliptin

Alogliptin

DPP -4 Inhibitors (“gliptins”)

Mechanism of Action

Endogenous incretins, GLP-1 and GIP have a half life **only 1-2 minutes**

Dipeptidyl peptidase 4 (DPP4) is the enzyme that inactivates GLP 1 & GIP

So if you Inhibit DDP 4 – you will have an increase in the amount of active GLP1, GIP

(What is GIP? **Glucose-dependent insulintropic peptide**

secreted by intestinal K cells in response to nutrient ingestion.

GIP helps to stimulate glucose dependent insulin secretion for the beta cells)

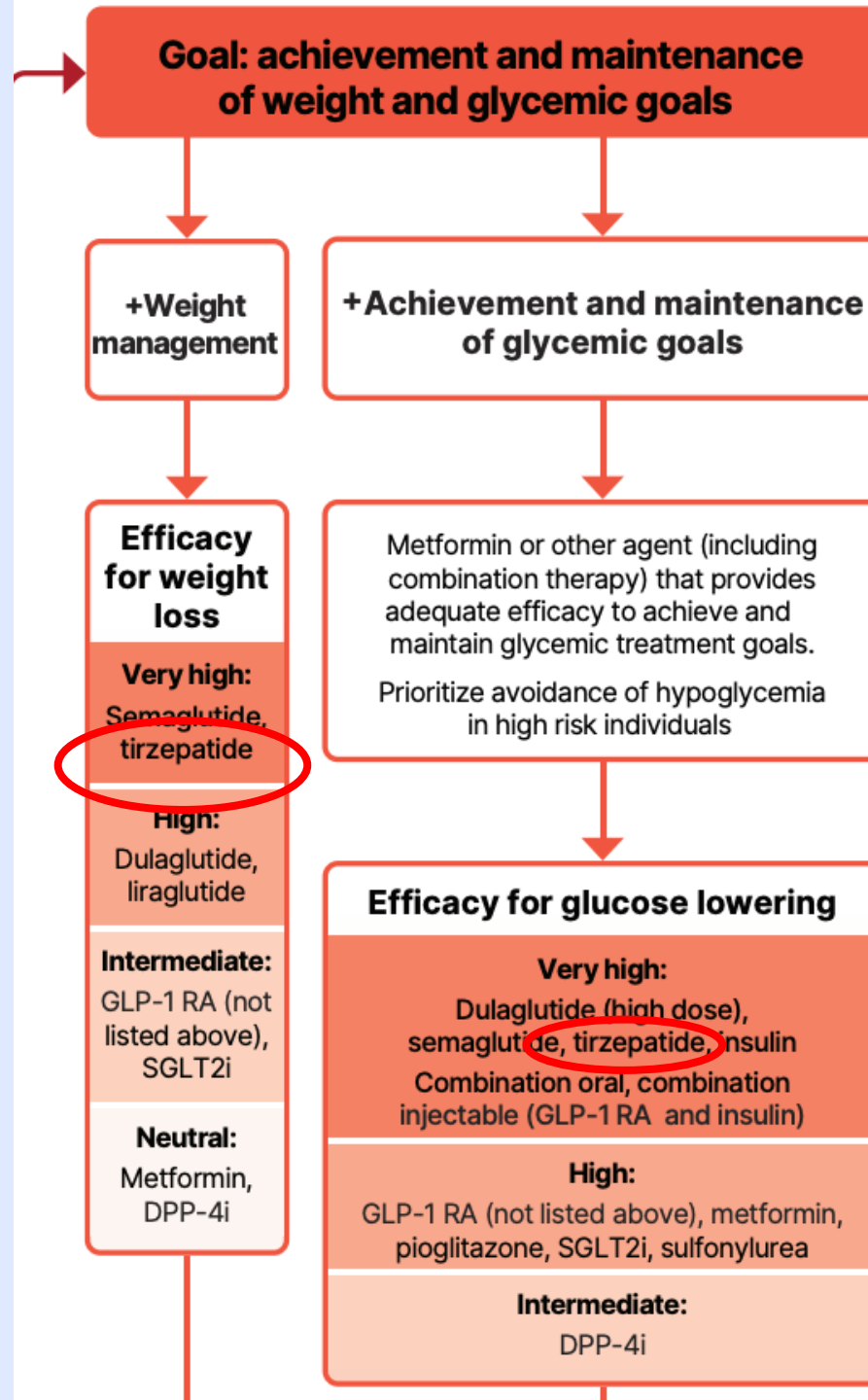
DPP-4 Inhibitors

Metabolism/ Adverse effects

Most are excreted unchanged renally. One (linagliptin) undergoes hepatic metabolism with biliary/ fecal excretion

Side effects are minimal with this class

- Commonly reported side effects include headache, nasopharyngitis, and upper respiratory tract infection.
- In postmarketing reports, DPP-4 inhibitors have been associated with hypersensitivity reactions, including anaphylaxis, angioedema, and blistering skin conditions, including Stevens-Johnson syndrome.
- Acute pancreatitis has been reported in association with DPP-4 inhibitors, they should not be initiated in a patient with a history of pancreatitis.



Dual GLP1RA and GIP-RA “Twincretin” Class-

Tirzepatide (FDA approval May 2022) (injected weekly)

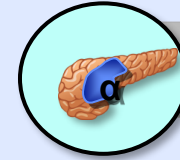
- Up to 2.58 % A1c lowering (with use of highest dose)
- Weight reduction up to approx. 15-20%
- Not associated with hypoglycemia
- ASCVD data- noninferior to dulaglutide

Mechanism of Action : selectively binds to and activates **both the GIP and GLP-1 receptors**, the targets for native GIP and GLP-1

Rarely Used agents for Type 2 Diabetes

- Amylin Analog -Pramlintide
- Alpha Glucosidase Inhibitors
- Bromocriptine
- Colesevelam

Amylin Analog (Pramlintide):



Glucagon
secretion

Amylin is a peptide co-secreted with insulin from beta cells. Pramlintide is an analog of amylin with the same effects (somewhat similar to GLP1RA)

- Inhibits glucagon secretion
- Slows gastric emptying
- Works in the CNS to promote satiety

Side Effect: Nausea (20% of patients)

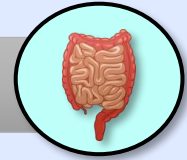
Clinical Use

- Approved for use in insulin treated type 2 diabetes, injected with mealtime insulin
- Lowers A1C 0.5-0.6%
- Mild reductions in weight (0.5 – 1.5kg)

Majority of patients now use GLP1RA!

Alpha Glucosidase Inhibitors

Glucose
uptake



- Mechanism: Slows gut absorption of Glucose by inhibiting intestinal alpha glucosidase (Reduces rate of polysaccharide → monosaccharide digestion in proximal small intestine)
- Not often used due to Side Effects: diarrhea, bloating, flatulence (delay in the absorption of carbohydrates causes more complex carbohydrates to appear in the colon where enteric bacteria break them down generating gaseous by-products)
- Only lower A1C by about 0.5%.
- Dosing is three times daily with the start of each main meal

Acarbose

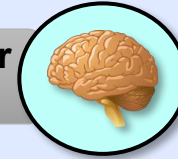
Miglitol

Colesevelam

- This agent is a bile acid sequestrant originally approved by the FDA to lower serum cholesterol levels
 - It was approved in 2008 for the treatment of type 2 diabetes
- The mechanism by which it lowers glucose is unknown
- Lowers A1C by only about 0.5%
- Major side effect is gastrointestinal – bloating, gas and diarrhea

Bromocriptine

Neurotransmitter
function



- This is a dopamine agonist
- Precise mechanism of action is not known, but it appears to work on the hypothalamic neurotransmitter activities that induce and maintain the glucose resistant state of diabetes.
- Very weak glucose lowering effect: lowers A1C about 0.4 – 0.5%
- Major side effects are nausea, vomiting, fatigue and dizziness.

Bottom line:

A patient-centered approach should be used to guide the choice of pharmacologic agents

Considerations include co-morbidities, glucose lowering efficacy, impact on weight, hypoglycemia risk, potential side effects, delivery method (oral versus injectable), cost, and patient preferences.

Extra-glycemic Benefits (cardiac/ renal) shown in Randomized Control Trials (by drug class) (note this means individual agents with proven benefit (RCT) from each class

ASCVD (atherosclerotic cardiovascular disease)

- GLP1RA
- SGLT2i

CHF (congestive heart failure)

- SGLT2i
- (Select GLP1RA/ GLP1-GIPRA with proven benefit in symptomatic HFpEF and obesity)

CKD (chronic kidney disease)

- SGLT2i
- Select GLP1RA

Hypoglycemia

No Hypoglycemia

- Metformin
- GLP1RA
- Twincretin GLP1RA/GIPRA
- SGLT2inhibitors
- DPP4 inhibitors
- TZD's

Hypoglycemia risk

- Sulfonylureas
- Meglitinides
- Insulin

Weight

Weight loss Benefit

- Strongest: Twincretin, GLP1RA
- SGLT2i
- Metformin

Weight Gain noted

- Insulin
- Sulfonylureas
- Meglitinides
- Thiazolidenediones (TZD)

QUESTIONS?

Case 1:

The patient is a 52-year-old with a medical history significant for hypertension, obesity (BMI = 31) and hyperlipidemia who was told that they were a “borderline” diabetic 2 years ago at their last physical exam . You order routine labs and find all are normal except for a fasting blood sugar of 160 mg/dL on two occasions and a hemoglobin A1C of 7.4%. Your best initial course of therapy is:

Case 2:

- The patient is a 50-year-old with a past medical history significant for coronary artery disease status post myocardial infarction with a cardiac stent placed in the LAD, HTN, hyperlipidemia, obesity, and type 2 diabetes for 6 years.
- Physical exam is only remarkable for a BMI of 34 and BP is 140/88.
- Laboratory evaluation reveals an A1C of 9%. The rest of the laboratory evaluation is normal.
- They are currently taking metformin and glyburide for diabetes.
- What options are available to treat the diabetes?

Case 3:

The patient is a 70-year-old with a history of coronary artery disease, congestive heart failure, chronic kidney disease, hypertension, hyperlipidemia, and glaucoma who presents to you for evaluation and treatment of diabetes. They have non-proliferative diabetic retinopathy and neuropathy. They are taking glyburide 5 mg once daily and their sugars are not well controlled (running around 200). Their labs show a GFR of 35 and a hemoglobin A1C of 8.8%. On physical exam there is bilateral lower extremity edema. What therapeutic changes should we make?